

ORIGINAL ARTICLE

Towards privacy-preserving and real-time GNSS interference detection in ADS-B systems using CNN

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Abstract

The growing dependence on Global Navigation Satellite Systems (GNSS) based aviation systems has exposed vulnerabilities in Automatic Dependent Surveillance-Broadcast (ADS-B), such as susceptibility to interference, jamming, and spoofing attacks. Conventional detection methods often suffer from issues like high false alarm rates, latency issues, and poor adaptability in real-world scenarios. To address these concerns, the current research presents an AI-powered anomaly detection framework that combines Convolutional Neural Network (CNN)-based spatial feature extraction with Gated Recurrent Unit (GRU) based temporal sequence learning, for enhanced real-time detection of GNSS interference in the aviation industry. The proposed CNN-GRU-Based Anomaly Detection and Flight Trajectory Prediction (CNN-GRU ADFTP) takes ADS-B flight telemetry data for training, fusing critical aircraft parameters like latitude, longitude, velocity, altitude, and vertical rate, and demonstrates better accuracy in anomaly detection with fewer false positives compared to traditional machine learning approaches. Geospatial anomaly evaluations also unveil areas of high-risk airspace, highlighting the need for ongoing aviation security monitoring. Although the promising results provided by the suggested framework, challenges like class imbalance and scalability issues persist. Future research will focus on refining model sensitivity, integrating external flight metadata and weather factors, and optimizing edge computing applications for enhancing low-latency processing capabilities in densely populated airspace environments. This study adds to the development of secure and resilient air traffic monitoring systems, thus enhancing aviation safety amidst emerging cyber threats.

Keywords ADS-B security · GNSS · Anomaly detection · Aviation security

1 Introduction

The introduction of Automatic Dependent Surveillance-Broadcast (ADS-B) systems has significantly enhanced air traffic control efficiency by providing real-time, accurate aircraft position data. As a core component of modern aviation infrastructure, ADS-B improves situational awareness and optimizes airspace management. However, despite advancements in security measures, critical vulnerabilities persist, exposing ADS-B systems to threats such as jamming, which disrupts communication channels, and spoofing, where unauthorized entities



intercept and manipulate ADS-B messages. These security breaches pose severe risks, potentially compromising air traffic management and endangering pilots and passengers.

With the growing adoption of ADS-B technology in the aviation industry, concerns over network security vulnerabilities have intensified. These weaknesses can undermine the reliability of aircraft surveillance systems, necessitating proactive security measures to detect and mitigate threats effectively [1]. Current approaches include upper-layer security mechanisms, such as strong authentication and encryption, to safeguard communications, and physical-layer security strategies aimed at preventing unauthorized access to ADS-B infrastructure [2–4].

Moreover, Global Navigation Satellite Systems (GNSS), including GPS and GLONASS, play a crucial role in ADS-B functionality by providing precise positioning and navigation services. However, GNSS signals are inherently susceptible to jamming and spoofing attacks, which can distort aircraft location data and lead to navigation failures. In response, next-generation GNSS solutions are being developed in collaboration between the European Union and China, focusing on enhancing interoperability while addressing existing security gaps.

The ADS-B system remains vulnerable to a range of cyber threats, including network intrusions, message tampering, and data corruption [3]. One approach to detecting spoofing involves analyzing signal propagation time between transmitters and receivers. Additionally, interference in GNSS signals can degrade positioning accuracy, necessitating robust verification mechanisms for ADS-B data [4]. Given the increasing sophistication of cyber threats, multi-layered security frameworks are essential for protecting mission-critical aviation systems [5]. The rise in state-sponsored cyber warfare and covert cyber operations has further amplified risks to Supervisory Control and Data Acquisition (SCADA) and air traffic control networks [6].

To address these challenges, the integration of Artificial Intelligence (AI) specifically Machine Learning (ML) and Deep Learning (DL) techniques offers a promising solution for enhancing ADS-B security. This paper introduces a novel AI-driven framework that leverages advanced AI models to detect and mitigate GNSS interference in ADS-B systems. The proposed approach incorporates hybrid AI techniques to analyze aviation safety data, predict potential security breaches, and implement automated risk-mitigation measures.

Despite extensive research on ADS-B security, there remains a significant gap in the application of AI-based methodologies for threat detection and response. This study aims to bridge that gap by developing an intelligent security system that utilizes AI-driven threat analysis and mitigation techniques. The findings of this research will contribute to enhancing ADS-B security frameworks, strengthening aviation cybersecurity, and ensuring the resilience and reliability of modern aircraft surveillance systems. “Privacy-preserving” in this research denotes the architectural design aim of minimizing visibility of operational telemetry data, mainly with inference mechanisms being performed at a more localized level and not at a centralized level. Though not implemented yet, some privacy-enhancing technologies (such as encryption or differential privacy mechanisms) are important future paths for improvement of this framework.

This research introduces an AI-driven framework for real-time GNSS interference detection in ADS-B systems, addressing critical aviation security challenges. The key contributions of this study include:

1. Developed a CNN-based model for detecting GNSS interference in real time, improving situational awareness and reducing false positives.
2. Implemented CNN models to accurately distinguish interference from authentic GNSS signals, ensuring low-latency processing for aviation security.
3. Integrated AI-driven predictive analytics to detect cyber threats like jamming and spoofing, strengthening system resilience.
4. Deployed lightweight AI models optimized for edge computing, ensuring efficient processing in high-traffic airspace conditions.
5. Provided higher accuracy, faster detection, and fewer false alarms compared to traditional methods.

The structure of this research is outlined as follows: Sect. 2 presents an overview of contextual background and underlying principles. The relevance and contribution of research in the applicable literature is highlighted in Sect. 3. The established paradigm is evaluated in Sect. 4. An explanation and interpretation of experiment-based

findings are discussed in Sect. 5. The research concludes in Sect. 6 by providing possible avenues for future research.

2 Background and basic concepts

Section 2.1 provides a brief overview of Security in Aviation. Section 2.2 provides Integration of Data and ADS-B. Section 2.3 focuses on Satellite Systems and Global Navigation Satellite Systems., Sect. 2.4 discusses the impact of Machine Learning and Deep Learning in the field of Aviation. Lastly, presents an overview about the Convolutional Neural Networks (CNNs).

2.1 Security in aviation

Safety and security have traditionally been separated into aviation. While the former addresses risks associated with aviation activities, the latter protects civil aviation from acts of unlawful interference. While national and international guidance exists on the management of risk for organizations with hazardous aviation operations, the appropriate application of established assessment techniques, both quantitative and qualitative, is critical for both areas. Incorrect identification of risks and the determination of a negative outcome can severely impact the level of protection, and the investments required to achieve it. The empirical example and data presented come from safety risk assessments in helicopter EMS flights. These flights use advanced automated flight procedures in obstacle-rich environments under low visibility conditions and are therefore a safety concern on the one hand. On the other hand, security is analysed whenever helicopter EMS flights are operated in adverse weather conditions, where the only navigation source is signals from the global navigation satellite constellation. Traditional security risk assessment (Wipf in Aviation risk and safety management, Springer, p 108, 1) under these circumstances only takes into account human performance factors under conditions of technical failure. However, security analysis should address all forms of jamming, spoofing, counterfeiting, and negative impact on receiver performance [7].

2.2 Integration of data and ADS-B

The Automatic Dependent Surveillance-Broadcast (ADS-B) system is instrumental in information exchange and ensures aircraft-airspace and aircraft-facility intercommunication. As a result, it gives a rich dataset related to aircraft movements, signal broadcasts, and sensor readings. The system is vital for jamming and interception threats in modern avionic systems while at the same time improving the reliability and accuracy of air traffic control. The structure of the ADS-B system includes interactivity in two main forms: (a) aircraft-aircraft and (b) aircraft-airport station. The information produced in ADS-B includes extensive information related to aircraft, signals, and sensors and related parameters. The information transmitted in ADS-B includes static and dynamic parameters. The static parameters include information related to planned arrivals and departures at terminals, aircraft identification or identification signals, and planned flight routes. The dynamic parameters are marked by four-dimensional (4D) characteristics, including time stamps, spatial coordinates (latitude, longitude, and height), velocity, and planned flight paths [4].

2.3 Satellite-assisted and global navigation satellite systems

Global Navigation Satellite Systems (GNSS) are constellations of emitting satellites, thus providing credible navigation service available at any place on Earth's surface at any time and in any conditions. As of 2010, navigation systems based on satellites have been in place and are managed by the United States Department of Defense and Russia. The Russian Federation maintains the Global Orbiting Navigation System (GLONASS) while the United States maintains the Global Positioning System (GPS). As of 2010, plans for the European Union and China to

have respective GNSS had commenced. The European Union system is known as Galileo while the Chinese system is known as Compass. The goal for each is to be independent and functioning in place, and while looking to expand usage beyond respective countries, especially in the Chinese context.

At present, there are 31 functioning satellites in the constellation, and they are allocated in a systematic manner across six circular orbits. Each of them is occupied by at least five satellites. The angular difference in position for each orbit is 60° , and they are tilted at an angle of 55° relative to the equatorial plane. The nearly circular and non-geostationary tracks are traveled in about 11 h and 58 min in one entire circle. On average, at least three GPS satellites are expected to be in sight at nearly all locations on Earth, but for precise and uninterrupted location calculation, four satellites are required [8].

The research explores the functions of operational and developmental Global Navigation Satellite Systems (GNSS) in as much as they can provide worldwide cover across conditions. The major functions are ensuring interoperability, ensuring autonomy, and improving complementarity. It is expected in the future that most aircraft would be equipped with receivers that have inter-constellation interface capability in demodulating signals from multiple GNSS for augmenting and improving performance. The paper cited in [9] presents a thorough explanation and elucidation of the functioning mechanisms for these systems, as well as a close look at some modern-day challenges.

2.4 The role of machine learning and deep learning for the aviation industry

Despite the serious challenges encountered, the wide presence of abundant data and computing resources enabled advancements and the implementation of artificial intelligence techniques in most areas of study. The paper presents a new safety assurance framework tailored for systems of the aviation industry. The model proposed improves predictability with respect to the occurrence likelihood of the hazard and the severity of the incident within the context of the aviation industry by using machine learning and deep learning techniques. The macro performance of the model in most cases improves by the use of diversity metrics for the main classifiers. The model development and validation utilize the dataset of the Aviation Safety Reporting System for twelve years of duration. To test the performance of the model, precision, recall, and the F-score are used to show the model superiority over [10].

Machine learning is essential in the verification of safety measures and predictive maintenance strategy implementation. Predictive maintenance has gained prominence in the business world because it allows for the anticipation of hardware failure and the proactive scheduling of repairs. Predictive maintenance uses automated data analysis to continuously track the status of hardware components throughout their working life, often recorded as time series data and event logs. The main challenge is the analysis of large volumes of historical observational data to create accurate forecasting models that can effectively predict potential future failures. This method uses regression methods in combination with machine learning algorithms, namely Gaussian Mixture Models, to assess risk factors. It provides a scalable solution that performs well in clustered environments for training and testing purposes [3].

2.5 Convolutional neural networks (CNNs)

Convolutional Neural Networks (CNNs) constitute a specific category of deep neural networks specifically designed for analyzing visual information, making them invaluable in the area of computer vision. The networks feature convolutional layers that use filters, which scrutinize images in the input, enabling detection of important features such as edges, textures, and contours. Pooling layers help minimize feature map dimension, thus making the networks more computationally efficient while increasing resistance against changes in the input information. The introduction of nonlinearity by activation functions enables CNNs to distinguish subtle patterns, while the application of dense layers at the output assists in the realization of precise predictions. The ability of CNNs to learn features autonomously from images eliminates the need for feature engineering by hand, thus increasing

their effectiveness in applications like image recognition, object detection, and image segmentation. In addition, their efficient implementation helps with real-time applications such as video processing in surveillance and activity detection, medical images in disease diagnosis, and guidance in self-driving cars. The resilience of CNNs in tolerating changes in object location and orientation also supports their usefulness in applications such as automated diagnosis, face detection, and satellite image processing [11].

3 Related works

The application of deep learning (DL) and machine learning (ML) has significantly impacted many fields, such as computer vision, natural language processing (NLP), detection in radiotelephony signals, and GNSS security. In 2016, researchers devised several activation functions for neural networks, such as ReLUs, Leaky ReLUs, Parametric Rectified Linear Units (PReLU), and Randomized Leaky ReLUs, in order to counteract problems like bias shift and improve resistance against noise. The introduction of Exponential Linear Units (ELUs) further improved learning efficiency, leading to increased model precision in visual datasets [2].

By 2017, deep learning techniques had surpassed traditional machine learning techniques in the areas of image classification, speech recognition, and radio signal classification. Advances in neural network design, optimization via stochastic gradient descent, and acceleration using graphics processing units drastically improved the efficiency and scalability of the models. However, issues like data transferability and gaps between simulated and real-world accuracy remained, highlighting the essentiality of transfer learning for real-world applicability [12].

In 2018, CNNs were applied for the first time to recognize and estimate sea ice concentration by analyzing TechDemoSat-1 (TDS-1) delay-Doppler maps (DDMs). The method outperformed traditional neural network-based methods, with better accuracy, efficiency, and sensitivity to changes in data structures [13]. Similarly, ML models were applied in the processing of Global Navigation Satellite System (GNSS) signals for detecting ionospheric scintillations, which cause satellite navigation system disruptions. The models enhanced detection accuracy while reducing false alarms, thus becoming vital for monitoring space weather events and performing atmospheric remote sensing [14].

By 2019, the advent of GPS spoofing and meaconing attacks created serious security issues for Unmanned Aerial Systems (UAS) since these attacks undermined both the accuracy of localization and time synchronization. Traditional countermeasures relied on auxiliary peak tracking, phase delay estimation, and spatial processing, which required modifications to the current GPS infrastructure. In response to this problem, a supervised machine learning approach based on artificial neural networks (ANNs) was proposed to detect spoofing events by analyzing the features of GPS signals, such as carrier phase, Doppler shift, pseudo-range, and signal-to-noise ratio (SNR). This method did not require infrastructure changes and could be implemented on low-power microcontrollers, making it an affordable solution for autonomous systems that use GPS [15].

In 2021, AI-driven GNSS security and interference detection gained momentum. Airbus introduced a deep learning-based classification model to detect Evil Waveforms (EWF) in GNSS signals, distinguishing multipath effects from deliberate interference. The method reduced reliance on expert-driven analysis and demonstrated the feasibility of DNN-based interference detection [21]. Meanwhile, GNSS navigation research expanded to include multi-layer perceptron (MLP) neural networks for spoofing detection, utilizing signal quality monitoring and distortion measurement metrics to improve jamming and spoofing resilience [16].

Additional advancements in deep learning-based beamforming techniques have improved the precision of LTE cellular positioning by reducing root mean square error (RMSE) rates of signal estimation [10]. Additionally, machine learning approaches were applied to Global Navigation Satellite System (GNSS) signal acquisition, employing modified Cross Ambiguity Functions (CAF) and deep neural networks (DNNs) to improve resilience to non-Gaussian noise. Experimental results validated the effectiveness of Multi-Layer Perceptron (MLP) and Convolutional Neural Network (CNN)-based architectures for GNSS detection [11].

Between 2022 and 2024, the development of deep learning models continued to be applied in GNSS multipath mitigation, spoofing detection, and satellite security enhancement. A CNN-based divide-and-conquer approach was proposed to facilitate parallelized GNSS signal acquisition, which led to better efficiency [17]. At the same time, DNN-based correlation models replaced traditional GNSS multipath mitigation techniques, demonstrating better tracking accuracy in complex signal environments [13].

By 2024, researchers reported more advanced deep learning architectures that can detect intentional attacks against Global Navigation Satellite Systems (GNSS), such as jamming, meaconing, and spoofing. A decision fusion-based methodology that combined support vector machines (SVMs), probabilistic neural networks (PNNs), and decision trees (DTs) was presented, aiming at enhanced detection accuracy [15]. In addition, a sparse autoencoder-based model was built to detect aerial spoofing threats against Low-Earth Orbit (LEO) satellite communications networks. While this method achieved high detection accuracy, its implementation was restricted by drone altitude regulations and was never tested in urban areas [18].

Overall, deep learning has altered the paradigm in GNSS security, signal classification, and signal processing methodologies. Further studies should be aimed at overcoming challenges in real-time implementation, adversarial robustness, scalability, and privacy concerns in order to further improve AI-driven solutions for secure and reliable navigation systems. Table 1 presents a comparison of related work.

Regardless of the advancements in Automatic Dependent Surveillance–Broadcast (ADS-B) security features and interference detection in Global Navigation Satellite Systems (GNSS), existing approaches face difficulties with real-time precision, flexibility, and computational complexity. Traditional methods rely on rule-based filtering, statistical anomaly detection, and simple machine learning, which often produce high false positive rates and long response times. In addition, most existing solutions lack deep learning architectures for feature extraction automation and real-time threat categorization. While previous studies aimed at GNSS spoofing detection through signal propagation assessment and physical-layer protection approaches, such methods entail hardware alterations and suffer from limited generalizability across various scenarios. Finally, depending on central computing platforms limits scalability and makes such approaches unfeasible in areas with high air traffic density [19].

Addressing the detected shortcomings, this research proposes a novel AI-driven framework leveraging Convolutional Neural Networks (CNNs) for real-time detection of GNSS interference in ADS-B systems. In contrast to traditional approaches, CNNs have the ability to automatically learn spatial and temporal features inherent in GNSS signals, thus significantly enhancing detection accuracy and reducing false positive events. The architecture is optimized for edge computing, enabling low-latency processing even under heavy airspace conditions. In addition, the use of AI-augmented predictive analytics enables the provision of timely warnings regarding detected threats, hence mitigating risks of jamming and spoofing before they can affect aviation security. By consolidating lightweight deep learning frameworks and scalable edge processing capabilities, the suggested approach offers an integrated, real-time security solution for modern air traffic surveillance and navigation systems [15].

4 Proposed CNN-GRU-based anomaly detection and flight trajectory prediction (CNN-GRU ADFTP)

The application of artificial intelligence in the aviation industry is critical in the prediction of system malfunction, optimization of maintenance process, and ensuring flight safety. Using real-time monitoring, advanced feature extraction techniques, AI-based classification, and trajectory prediction, this system detects and prevents anomalies in aircraft flight activity. Using Convolutional Neural Networks (CNNs) greatly enhances the accuracy and effectiveness in anomaly detection by learning spatial and temporal features autonomously from ADS-B.

Figure 1 illustrates a systematic method for flight path forecasting with Automatic Dependent Surveillance–Broadcast (ADS-B) information complemented by classification methods using convolutional neural networks

Table 1 A comparison of related works

References	Year	Contributions	AI	Advantages	Limitations
[1]	2023	This paper provides an in-depth review and discussion of the present condition of physical layer security (PLS) in satellite communication. It categorizes primary applications and discusses enhancements in security as well as ongoing research difficulties	No	The study addresses the significance of PLS in ensuring the security of satellite communication, especially in situations with limited resources such as IoT and satellite networks. It provides guidance for the advancement of more secure systems	The review focuses limited on PLS approaches and does not address practical implementation aspects
[2]	2016	Developed activation functions (ReLU, Leaky ReLU, Parametric ReLU) to improve neural network learning efficiency	Yes	Enhances neural network training stability and resilience to noise	Dying ReLU problem can cause inactive neurons, degrading model performance
[3]	2017	Deep learning has replaced machine learning in computer vision, voice, and natural language processing, improving sensitivity and accuracy in radio signal identification. It can learn large neural network models with high parameters, enhancing real data capture and labeling	Yes	Achieves high accuracy in real-time radio signal processing	Requires large labeled datasets and significant computational resources
[4]	2018	The study employs convolutional neural networks (CNNs) for the first time to detect and estimate sea ice concentration in TechDemoSat-1 delay-Doppler maps, outperforming current methods	Yes	Achieves high accuracy in real-time radio signal processing	Requires large labeled datasets and significant computational resources
[5]	2018	The paper presents a machine learning-based method for detecting amplitude scintillation, outperforming current methods in accuracy and F-score. This method offers increased generalization, location-independence, and avoids false alarms due to multipath reflections, making it crucial for space weather applications and atmospheric remote sensing	Yes	This method offers increased generalization, location-independence, and avoids false alarms due to multipath reflections, making it crucial for space weather applications and atmospheric remote sensing	Limited robustness to sudden high-frequency interference
[6]	2019	The paper proposes a supervised machine learning method using artificial neural networks (NN) to detect GPS spoofing messages. The algorithm uses received GPS signals to determine the presence or absence of the attack, without modifying GPS infrastructure. The method can be implemented using microcontrollers or microcomputers in any GPS-driven autonomous system. The paper analyzes five features: satellite number, carrier phase, pseudo range, Doppler shift, and SNR, and evaluates the efficiencies of different neural networks with different hidden neurons	Yes	The paper analyzes five features: satellite number, carrier phase, pseudo range, Doppler shift, and SNR, and evaluates the efficiencies of different neural networks with different hidden neurons	Limited to specific GPS attack types and may not detect advanced spoofing techniques
[7]	2020	Airbus is using AI to detect evil waveforms (EWF) in GNSS signals, using deep neural networks to classify nominal, multipath, and EWF cases. The method reduces the need for dedicated expert systems and shows feasibility in EWF detection. The model's application is also explored for jamming and spoofing scenarios	Yes	The method reduces the need for dedicated expert systems and shows feasibility in EWF detection	Requires continuous model updates for new spoofing techniques
[8]	2020	Navigation research focuses on GNSS interference detection and mitigation, including jamming, multi-path, and spoofing. Security and monitoring are crucial, with techniques like pre-dispreading and post-dispreading, signal quality monitoring, and a new GPS spoofing detection method	No	Security and monitoring are crucial, with techniques like pre-dispreading and post-dispreading, signal quality monitoring, and a new GPS spoofing detection method	Lacks real-time response mechanisms for active interference mitigation

Table 1 (continued)

References	Year	Contributions	AI	Advantages	Limitations
[9]	2020	The paper introduces a gradient enhance method for deep neural networks, incorporating short circuits, which improves performance without high external data or computation costs, particularly in computer vision and natural language processing tasks	Yes	The paper introduces a gradient enhance method for deep neural networks, incorporating short circuits, which improves performance without high external data or computation costs, particularly in computer vision and natural language processing tasks	Performance gains vary based on network architecture and dataset properties
[10]	2020	The paper introduces a deep learning-based beam forming approach, enhancing accuracy for cellular LTE signals and reducing position root mean-squared error, compared to traditional SAN-based beam forming methods	Yes	Enhances signal accuracy in urban environments	High computational complexity and requires real-time processing power
[11]	2020	Signal acquisition in the Global Navigation Satellite System (GNSS) relies on the Cross Ambiguity Function (CAF) and Receiver Operating Characteristics (ROC). However, experiments show reality is more challenging than assumed. Modifying the CAF to be more robust to non-Gaussian behaviors has shown outstanding performance. This work proposes that efficient data-driven methods can learn complex behaviors, using NNs for detection in GNSS. MLP provides slightly better results due to the classification task	Yes	This work proposes that efficient data-driven methods can learn complex behaviors, using NNs for detection in GNSS. MLP provides slightly better results due to the classification task	Performance degrades in noisy, multipath-heavy environments
[20]	2020	The paper proposes using deep learning models to model multipath propagation effects on GNSS correlation outputs, replacing standard schemes with Neutral Network-based schemes. The trained model shows promising time-delay tracking and enhances robustness in challenging environments	Yes	The trained model shows promising time-delay tracking and enhances robustness in challenging environments	High computational demands and requires extensive training data
[21]	2022	The Global Navigation Satellite System (GNSS) faces threats from multipath propagation and interference-rich environments. A new method using data-driven Deep Neural Networks (DNN) models can mitigate these issues. This approach maintains tracking accuracy in multipath-rich environments, but requires a multi-correlation scheme, increasing computational costs. Future work should focus on complexity reduction and real data testing	Yes	This approach maintains tracking accuracy in multipath-rich environments	requires a multi-correlation scheme, increasing computational costs
[22]	2023	The Global Navigation Satellite System (GNSS) uses Deep Neural Networks (DNNs) for signal acquisition, focusing on Convolution Neural Networks (CNNs) as binary classifiers. A divide-and-conquer approach parallelizes computation using CNNs processing smaller CAF portions. Results show remarkable performance, opening the possibility for further GNSS problem advancements	Yes	A divide-and-conquer approach parallelizes computation using CNNs processing smaller CAF portions	Scalability issues when dealing with high-dimensional data
[23]	2024	The paper explores deep learning in GNSS receivers to detect intentional attacks like jamming and spoofing. It proposes a new approach using a deep neural network to improve satellite visibility and pseudo-range errors in urban areas. The method outperforms current methods, especially in moderate to high signal-to-noise ratios	Yes	The method outperforms current methods, especially in moderate to high signal-to-noise ratios	Fails in extreme low-SNR conditions and needs continuous training updates
[24]	2024	The paper introduces a new method for detecting aerial spoofing attacks on Low-Earth Orbit satellite communication systems, using a sparse Autoencoder. The solution outperforms competitors, but requires drone flight beyond 120 m altitude and urban environments	Yes	The open-source dataset collected contributes to further research on satellite security	The study cannot fly the drone beyond EU regulations' 120 m altitude and does not test in urban environments

Table 1 (continued)

References	Year	Contributions	AI	Advantages	Limitations
[25]	2025	AI for aviation safety and efficiency in addressing navigational challenges. It presents a hybrid model that combines the strengths of SVM and Gated Recursive Units (GRU) for anomaly detection	Yes	The study applied the DBSCAN (Density-Based Spatial Clustering for Applications with Noise) algorithm to analyze identified anomalies in flight patterns	Adaptability to emerging threats

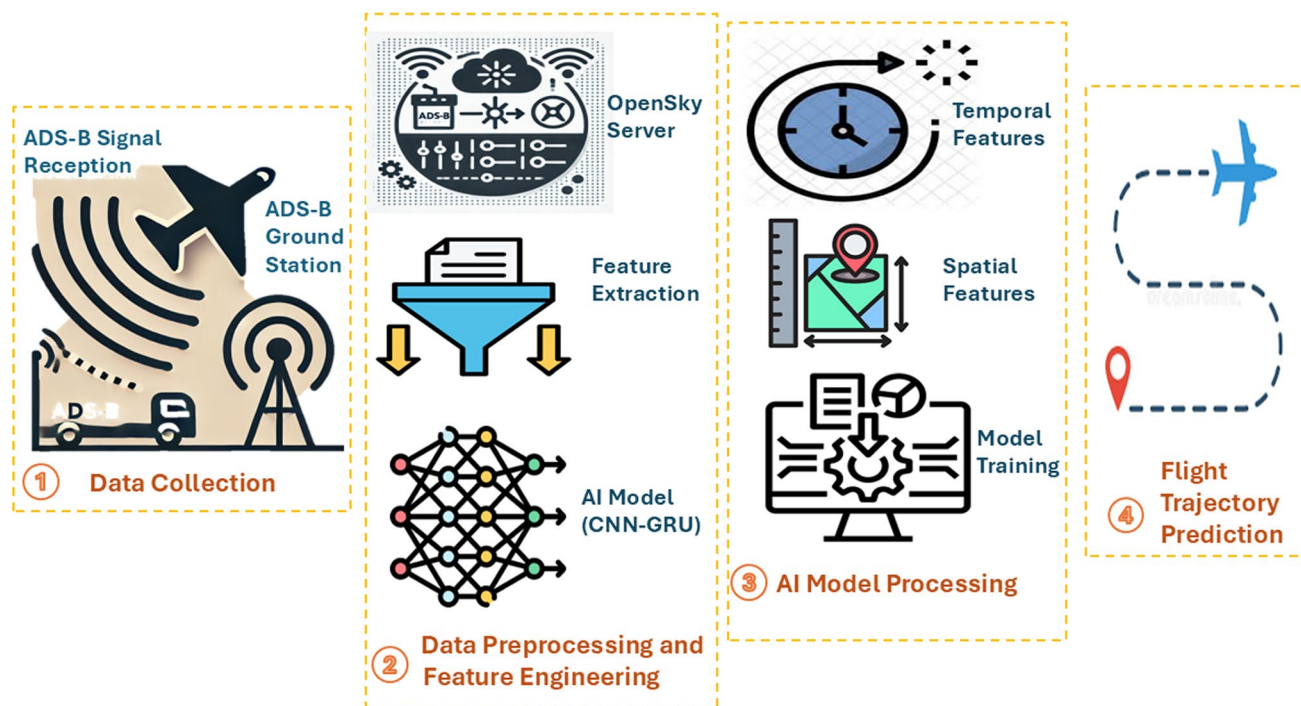


Fig. 1 Proposed (CNN-GRU ADFTP)

(CNNs). The model significantly promotes anomaly detection and control in flight tracks. Unlike earlier models, CNNs automatically derive important features from ADS-B signals, making the detection and categorization of unusual flight behaviors more precise. The framework includes the following steps:

4.1 Data acquisition (reception of ADS-B signals)

The process starts with the equipping of aircraft with ADS-B systems, which transmit real-time flight information that includes position, altitude, and speed to ground-based stations. The information is later collected and presented for further analysis. Algorithm 1 illustrates the method for gathering and arranging real-time ADS-B information for enhanced observation of flights and detection of anomalies.

4.2 Data preparation and feature creation

After collecting ADS-B data, this data is then stored in centralized databases, for example, the OpenSky Network, where the raw data is processed to extract spatial and temporal features, such as latitude, longitude, altitude, speed, and timestamps. These features are crucial in analyzing aircraft behavior. The suggested hybrid approach combines GRU and CNN for feature extraction and classification purposes. The GRU is specifically good at

modeling temporal relationships by analyzing sequential flight trajectories, while CNN is good at analyzing spatial data by recognizing subtle patterns and anomalies. The hybrid CNN-GRU model allows precise detection and classification of anomalies in flight operations. The suggested framework utilizes a hybrid architecture that integrates Convolutional Neural Networks (CNNs) and Gated Recurrent Units (GRUs), effectively harnessing the capabilities of CNNs for feature extraction alongside GRUs for the identification of temporal relationships within sequential data. The CNN components are tasked with detecting localized patterns in the incoming GNSS interference data, whereas the GRU components manage the dependencies at the sequence level, thereby enhancing the model's suitability for tasks involving anomaly detection in time-series data.

It takes as input GNSS interference signal sequences, where each sequence is expressed as a constant-length observation window. The first three layers of the network include convolutional layers, each followed by a max pool layer that allows for downsampling as well as the conservation of important spatial details. The first convolutional layer is followed by 32 filters, each (3×3) in dimension, with a stride of 1 and the use of 'same' padding. In the second convolutional layer, there are 64 such (3×3) filters, while the third contains 128 (3×3) filters. After each convolutional layer, there is the ReLU activation function, which adds non-linearity to the network, allowing the latter to learn complex patterns effectively.

After the convolutional layers, the output feature maps are passed through a flatten layer that converts the 2D feature maps into a single-dimensional vector. This vector becomes the input to a Bidirectional GRU layer, where the sequence is processed both forward as well as backward to capture long-range relationships. The GRU layer is comprised of 128 units, each activated by ReLU. A Dropout layer, rate=0.5, comes after the GRU layer to prevent overfitting. The Gated Recurrent Unit (GRU) layer output is then passed to a fully connected dense layer containing 256 units that are complemented by ReLU activation functions, allowing the model to make higher-order inferences regarding the features that were recognized. A sigmoid output layer consisting of a single neuron is finally used to classify the input sequence either as anomalous or normal, effectively performing a binary classification. Training is carried out with binary cross-entropy as the loss, which is suitable for binary classification problems. Optimizer used is the Adam optimizer, which is suitable to train deep networks as it automatically adjusts the learning rate during the training, set at 0.001. To prevent overfitting, the dense layer weights are regularized under the term L2 with a penalty term of 0.001, and early stopping is implemented to interrupt the training if the loss on the validation dataset is not improving over the last 10 epochs.

This framework balances feature extraction with sequential temporal modeling, offering an effective technique to identify infrequent GNSS interference anomalies in the ADS-B infrastructure. Merging the use of CNN layers with the full-fledged recurrent nature of GRU layers allows the framework to adequately perceive both spatial as well as temporal dimensions present within the interference signals, ensuring effective performance on actual-world anomaly identification tasks.

4.3 Processing and training of AI models

Extracted ADS-B features are categorized into two types: (i) Temporal Features: Represent time-related changes, such as fluctuations in speed, altitude, and heading. (ii) Spatial Features: Plot the aircraft's geographic location, tracking it through the different airspaces. The proposed AI model integrates GRU and CNN to efficiently analyze flight data. GRU specializes in capturing time-series patterns, transforming sequential data into high-dimensional feature vectors. These vectors are then processed by CNN layers, which extract spatial correlations and classify flight behaviors. By leveraging CNN's automatic feature extraction capabilities along with GRU's sequential analysis, the model improves anomaly detection accuracy and reduces false positives.

Following ADS-B data collection, the dataset was processed to reduce the class imbalance that existed between anomalous and normal flight behaviors. In a bid to offer the model a more balanced view of both classes during training, oversampling was done on the anomalous class. This technique involves increasing the instances of objects in the minority class such that the model is ensured of sufficient data for successful learning while at the same time reducing bias towards the majority class. It is thus that the CNN model is enabled to identify rare

anomalies more effectively, leading to increased proficiency in tasks of detecting anomalies. Future versions of the model will examine the use of SMOTE (Synthetic Minority Over-sampling Technique) for creating synthetic data points for the minority class, thus enhancing class balance even more. In addition, we incorporated weighted loss functions for enforcing higher penalty for the misclassifying of anomalous cases, thus ensuring that the model puts more focus on detecting these rare events. To improve the model's performance on anomalous instances even further, we modified the loss function while training by incorporating a weighted loss. This imposes greater punishment for wrongly classifying anomalies while enabling the model to focus more on learning rare anomalies rather than overfitting on the majority class.

4.4 Flight trajectory prediction

After training, the AI model is applied for real-time prediction of flight trajectories, thus ensuring optimization of flight routes, delay prediction, and safety enhancement through real-time detection of anomalies. Algorithm 1 explains the real-time prediction process, defining the process utilized by the CNN-GRU model in predicting flight trajectories and detecting anomalies. The proposed framework enhances feature extraction, improves classification accuracy, and ensures real-time anomaly detection in flight operations. This approach significantly advances aviation safety, air traffic management, and predictive maintenance strategies.

Algorithm 1. CNN-GRU-based anomaly detection and flight trajectory prediction (CNN-GRU ADFTP)

Input: Real-time state transmissions from aircraft $S_i(t)$.

Step 1: Data Collection and Structuring

1. Initialize ADS-B data collection from transponders.
2. Extract key flight parameters: $F = \{lat, lon, alt, vel, heading, time\}$.
3. Store collected data in OpenSky Network.
4. Structure dataset $\mathbf{D}$ for preprocessing.

Step 2: Data Preprocessing and Feature Engineering

1. Normalize numerical features using **Min-Max Scaling**:

$$\mathcal{X}' = \frac{\mathcal{X} - \mathcal{X}_{min}}{\mathcal{X}_{max} - \mathcal{X}_{min}} \dots\dots\dots (1)$$
2. Compute temporal features:

$$\Delta v = v_t - v_{t-1}, \Delta alt = alt_t - alt_{t-1} \dots\dots (2)$$
3. Compute spatial displacement:

$$d = \sqrt{(lat_2 - lat_1)^2 + (lon_2 - lon_1)^2} \dots\dots\dots (3)$$
4. Construct time-series sequences for AI model input:

$$\mathcal{X} = \{\mathcal{X}_{t-k}, \dots, \mathcal{X}_{t-1}, \mathcal{X}_t\} \quad k = \text{sequence length} \dots\dots\dots (4)$$
5. Return preprocessed dataset $\mathbf{D}'$.

Step 3: CNN-GRU Model Training

1. Feature extraction using CNN:
 - Apply convolution operation:

$$f_{cnn}(\mathcal{X}) = ReLU(W_c * \mathcal{X} + b_c) \dots\dots\dots (5)$$
 - Reduce dimensionality using max-pooling:

$$P = \max(f_{cnn}(\mathcal{X})) \dots\dots\dots (6)$$
2. Temporal pattern learning using GRU:

$$h_t = GRU(W_g \cdot P + U_g \cdot h_{t-1} + b_g) \dots\dots\dots (7)$$
3. Anomaly classification using dense layer:

$$\mathcal{Y} = \sigma(W_o \cdot h_t + b_o) \dots\dots\dots (8)$$
4. Train model using binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)) \dots\dots\dots (9)$$
5. Update model parameters using Adam optimization:

$$\theta = \theta - \eta \nabla_{\theta} \mathcal{L} \dots\dots\dots (10)$$
6. Return trained CNN-GRU model.

Step 4: Flight Trajectory Prediction and Anomaly Detection

1. Extract features from new ADS-B input $\mathcal{X}'$:

$$\mathcal{X}' = f_{cnn}(\mathcal{X}) + GRU(\mathcal{X}) \dots\dots\dots (11)$$
2. Predict next flight position:

$$\hat{\mathbf{x}} = (W_o \cdot h_t + b_o) \dots\dots\dots (12)$$
3. Compute anomaly score based on deviation threshold δ :

$$A = \begin{cases} 1, & \text{if } |\mathcal{X}' - \hat{\mathbf{x}}| > \delta \\ 0, & \text{otherwise} \end{cases} \dots\dots\dots (13)$$
4. Trigger alert if anomaly detected ($A=1$).
5. Return predicted trajectory $\hat{\mathbf{x}}$ and anomaly status $\mathcal{A}$.

Output: Predicted flight trajectory $\hat{\mathbf{x}}$ and anomaly status $\mathcal{A}$.



5 Experimental results

5.1 Environmental results

The experiments were conducted with the use of Google Colab, a cloud-based platform for Jupyter notebooks with the benefits of GPU and TPU acceleration, thus making the training and evaluation of deep learning models efficient. The choice to use the Colab environment was driven by the scalability, accessibility, and the ability to use the pre-installed machine learning libraries such as TensorFlow, Keras, and Scikit-learn. Additionally, Google Colab provides seamless integration with Google Drive, thus making it easy to store and access huge datasets. The

main dataset utilized in this research was obtained from flight telemetry recordings from the OpenSky Network (<https://opensky-network.org>). The dataset included key variables related to aircraft state, including latitude, longitude, velocity, altitude, heading, rate of climb/descent, and timestamps, which represented both normal and abnormal flight behaviors. For the purpose of reproducibility, the dataset was allocated 70% for the train dataset, 15% for the validation dataset, and the remaining 15% for the test dataset. This allocation ensured that the development processes were balanced with regard to evaluation. It further prevented data leakage.

5.2 Confusion matrix evaluation

The confusion matrix (Fig. 2) illustrates the performance of the anomaly detection model. From the updated results, we observe that 28,972 normal cases (True Negatives) were correctly identified, and 9,849 anomalous cases (True Positives) were successfully detected. However, 713 normal cases were incorrectly classified as anomalies (False Positives), and 838 anomalies were misclassified as normal (False Negatives). These results reflect the success of the model in identifying anomalies after the application of class rebalancing techniques, such as oversampling the anomalous class and weighted loss functions.

Confusion matrix indicates that although the issues related to class imbalance have been somewhat mitigated, significant opportunities remain to improve the reduction of False Positives and False Negatives. Continued efforts in model optimization, encompassing supplementary methods like SMOTE and training strategies aimed at anomalies, are essential for augmenting the model's resilience, particularly in accurately identifying infrequent anomalies with greater precision.

5.3 Class distribution

Figure 3 shows the distribution of the alerts in the database before oversampling, with the presence of an extremely skewed ratio of normal to anomalous flight events. Most records are in the class “False” and only a minor minority are “True” (the anomalies). Such skewed distribution can lead to biased performance on the model, with the classifier learning to label everything as the most common class. Resampling techniques or performance measures specifically for anomaly detection need to be adopted to provide fair evaluation.

Oversampling was used as a means to deal with the problem of class imbalance. Oversampling was performed on the minority class, referred to as anomalies in this problem. The oversampling factor was adjusted to ensure a balanced class ratio without having the minority class dominate. The resulting balanced set had a balanced ratio of normal to anomaly samples of approximately 3:1.

After the recalibration of alerts distribution across the dataset, the number of False alerts (normal instances) is 29,810, while the number of True alerts (anomalous instances) is captured as 10,562. An improvement is reflected

Fig. 2 Confusion matrix

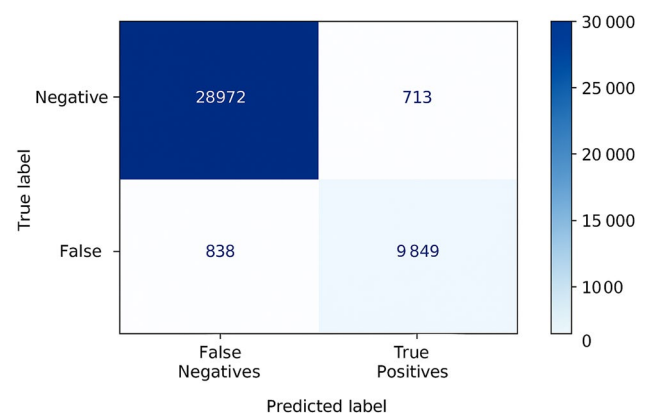
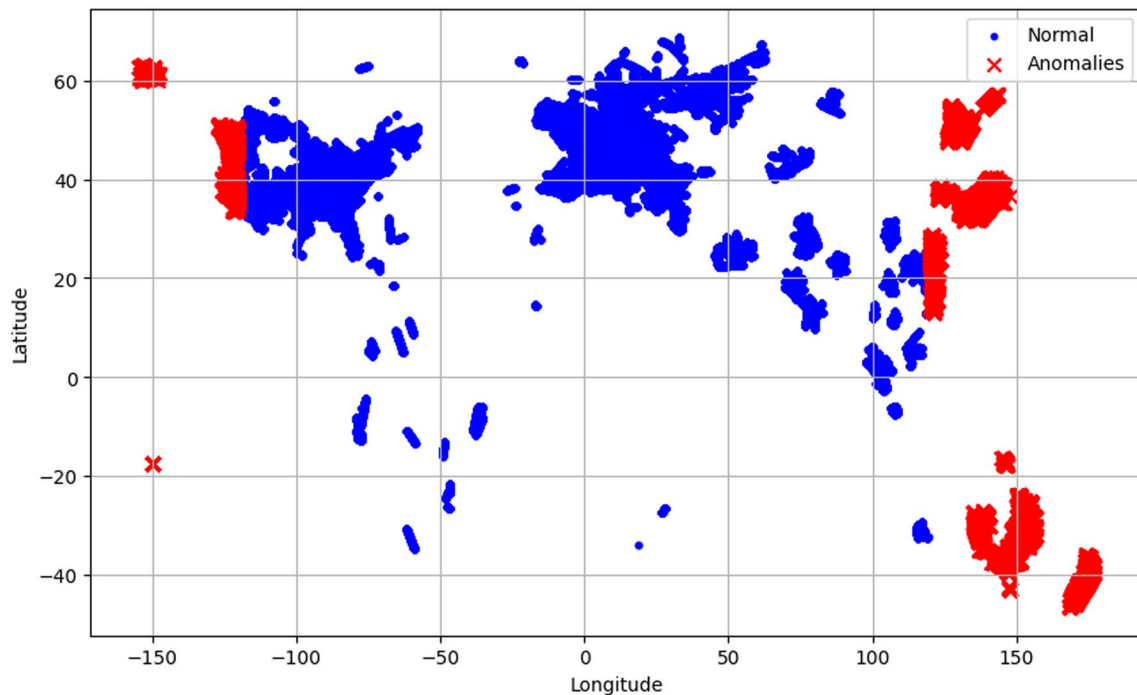
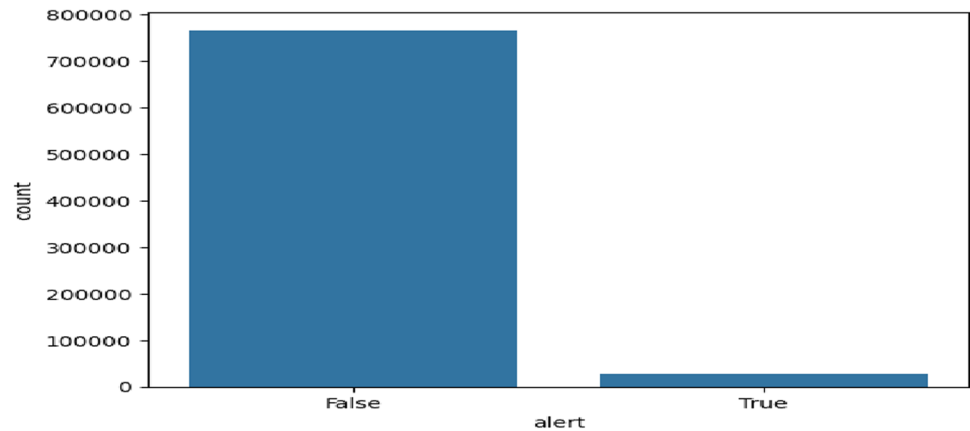


Fig. 3 Class distribution**Fig. 4** Latitude and longitude with anomalies highlighted

in the distribution as compared to the previous imbalance as shown in Fig. 3, where the algorithm strongly biased the instance classification as normal.

In spite of executing measures such as oversampling of rare instances to face the issue of class imbalance, the occurrence of False Positives (713) and False Negatives (838) points to future improvement opportunities. These indicate that, while the model has enhanced the effectiveness of catching anomalies, there still is a need to undertake further work so as to get the performance more balanced among both classes.

5.4 Geospatial anomaly mapping

Figure 4 illustrates the latitude–longitude graph with red representing anomalies and blue representing normal paths. There are anomalies at specific locations in the graph, which can be signs of localized disruptions, faulty

sensors, or intrusions into airspace. It benefits air traffic controllers and safety analysts because it enables them to identify dangerous hotspots and be able to improve monitoring processes.

5.5 Plane level anomalies

Figure 5 shows the total count for each plane according to ICAO24 code. There is an inference by the distribution that certain planes possess extremely high anomaly rates when compared to others, which can be referring to operational problems, faulty sensors, or navigational interference. It shows that intensive investigation into the most risky aircraft is necessary to avoid potential safety risks.

5.6 Alert and SPI flag comparison

Figure 6 shows the total alerted records and the SPI flags by type of anomaly. Most alerted records are type “False”, which verifies the model bias towards normal cases. However, some percentage of the actual alerted records and the SPI flags were successfully identified, meaning the model can identify some abnormal flight activities but needs to be optimized.

5.7 Anomaly velocity trends and predictions

Velocity fluctuation with respect to time, ARIMA-based forecasting, and anomaly detection are displayed in Fig. 7. A green line indicates the actual velocity and the blue line indicates ARIMA-based forecasting. Red marks indicate the anomaly detection and can be an indication of sudden spikes or bursts in velocity. They can be an indication of turbulence, sensor faults, or external noise, and need to be detected in real-time to avoid flight dangers.

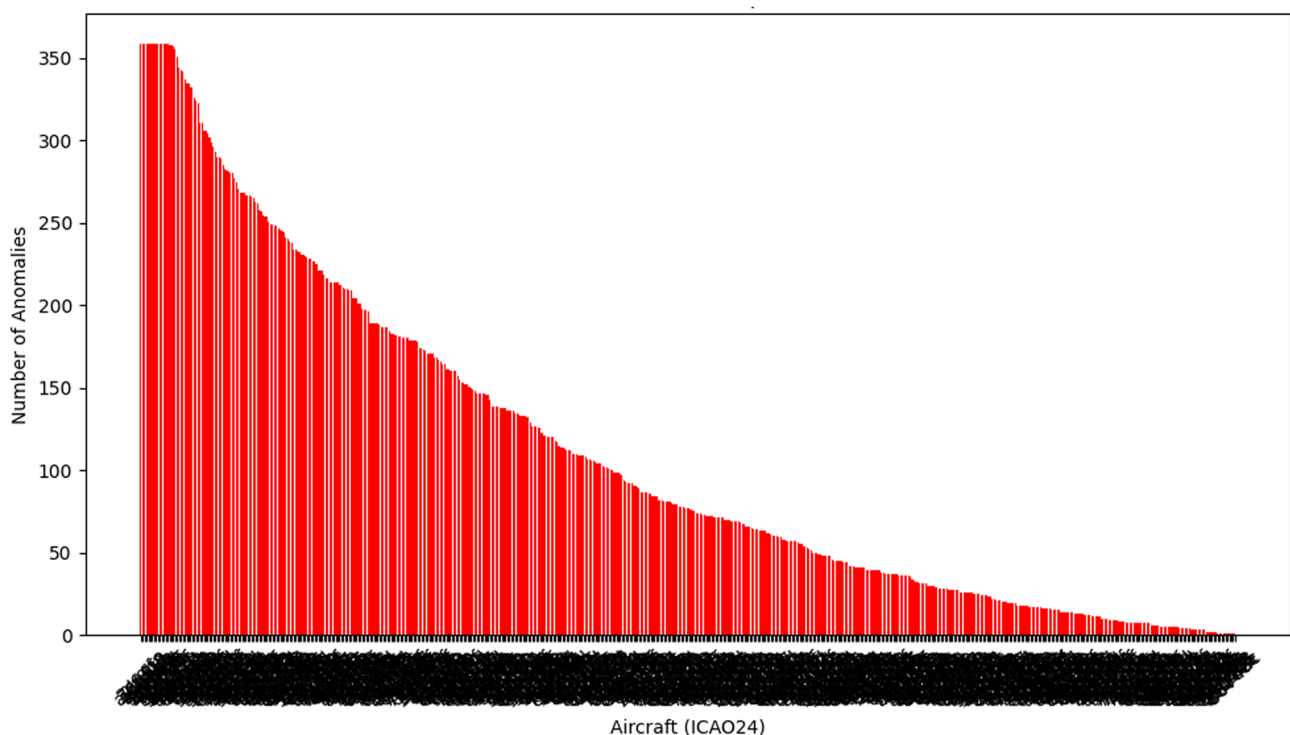


Fig. 5 Number of anomalies detected per aircraft (ICAO24)

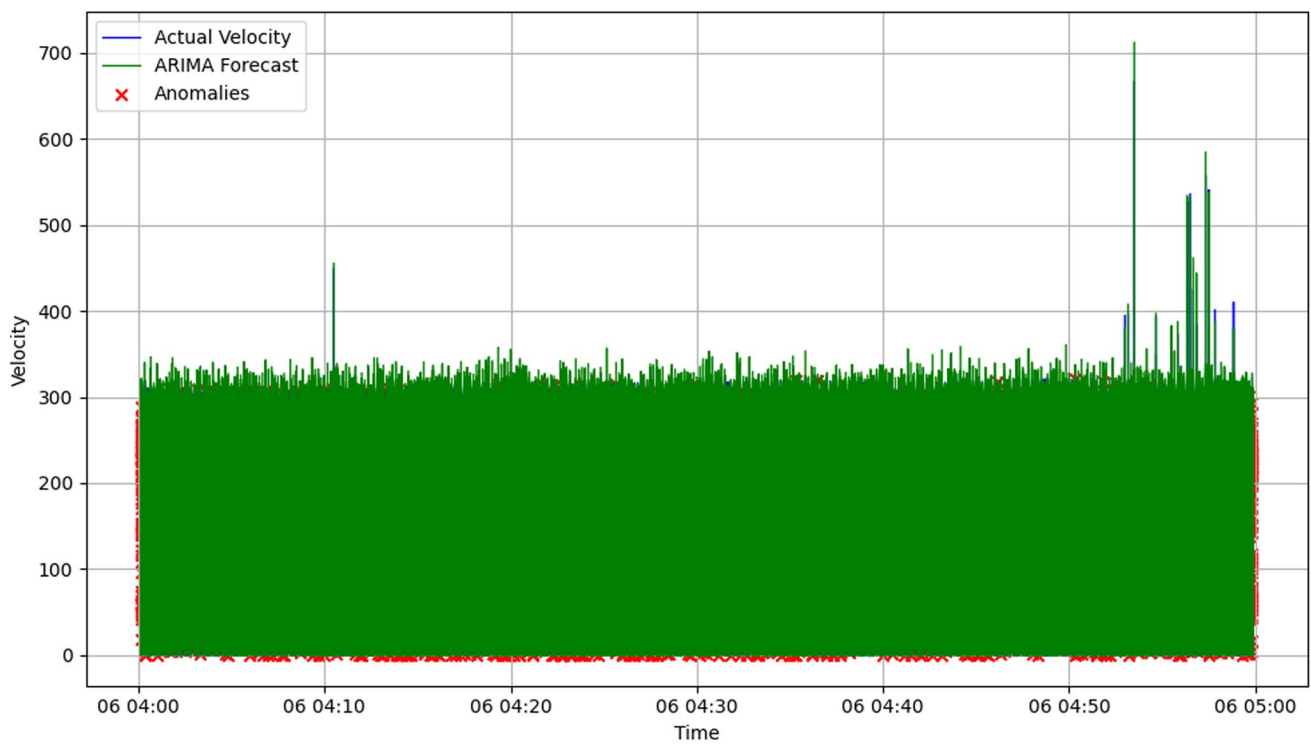
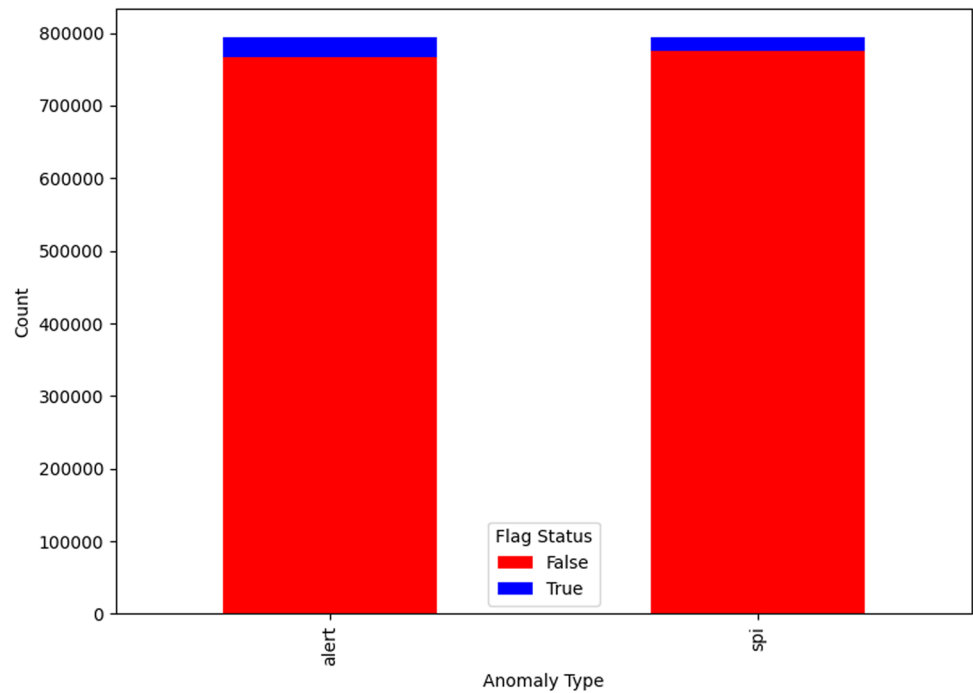
Fig. 6 Count of alerts and spi status by anomaly type**Fig. 7** Velocity over time with ARIMA forecast and anomalies

Fig. 8 Correlation heatmap

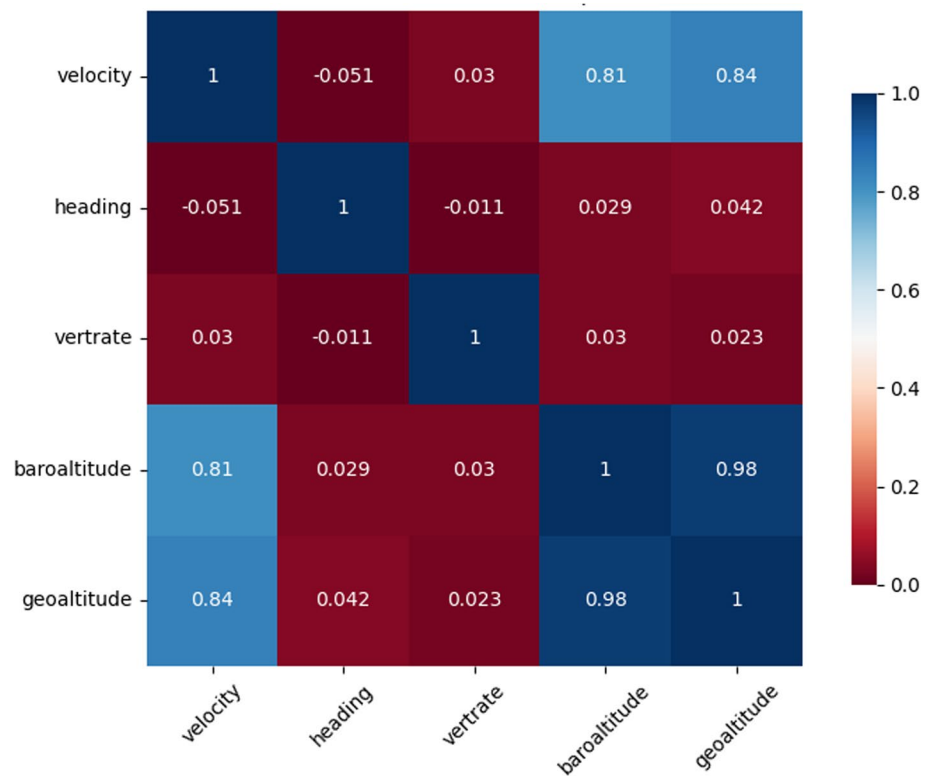
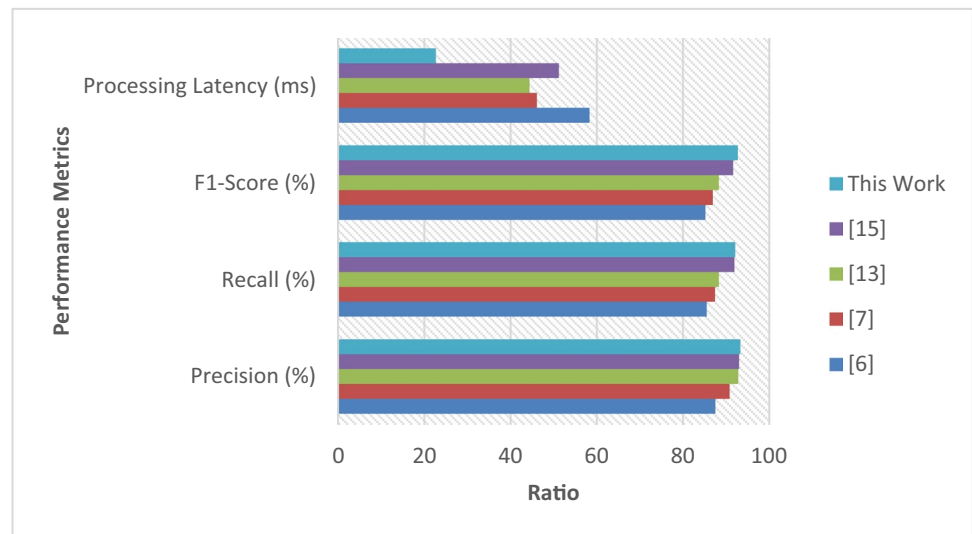


Fig. 9 Comparison with related work



5.8 Feature correlation analysis

Correlation heat map (Fig. 8) shows interdependencies among critical parameters such as velocity, heading, vertical rate, barometric height, and geometric height. High correlations among factors related to height suggest that height anomalies are likely to be a reliable indicator for potential navigation breakdowns. Noncorrelation among velocity and heading can also imply that certain anomalies comprise unexpected direction changes with no association with speed change (Fig. 9).

5.9 Comparison with related work

This section provides an experimental comparison of the performance achieved during the proposed CNN-GRU-based GNSS interference detection framework with five associated research studies. It specifies significant comparison metrics such as Detection Precision, Recall, F1-Score, Processing Latency, and Model Scalability, critically comparing these architectures: ANN-based GPS spoofing detection [6], DNN-based Evil Waveform (EWF) Detection [7], DNN-based multipath mitigation [13], and Deep learning for GNSS spoofing and jamming detection [15].

It achieves high performance across all metrics, such as a Precision of 93.3%, Recall of 92.1%, and F1-Score of 92.7%. This indicates that the framework is highly precise in anomaly prediction while ensuring the false positives as well as false negatives are kept to the minimum. These results are also shown in the comparison table below as the bar chart. To deal with the variation, the model was trained and tested on five separate runs. The F1-score showed a standard deviation of $\pm 0.3\%$, which was consistent across the testing sessions.

It surpasses all former efforts both in recall and precision, surpassing [6] and [7], as well as reaching very close to [15], which obtained 93% Precision and 91.9% Recall. Moreover, the framework of CNN-GRU excels because it boasts a very low processing latency of 22.7ms, significantly lower than conventional DNN-based ones, as we also indicate both in the table comparison as well as the bar chart comparison.

In processing latency, the CNN-GRU model performed best, achieving an inference time of 22.7ms, distant ahead of the other models presented, such as [6], which achieved a latency of 58.3ms, and [7], which required 46.1ms. Such low latency becomes very important in real-time applications related to aviation security, where the threat detection speed is paramount. It also exhibits scalability as it can be implemented on edge computing platforms where there is minimal utilization of the power as well as efficient anomaly detection functionality, thereby being a viable solution to the real-time monitoring of congested airspace.

To assess the real-time viability of our proposed framework, we have approximated the average inference time per sequence of inputs. Under the Google Colab environment configured with the Tesla T4 GPU-enabled setup, our CNN-GRU methodology gives us an average inference time of ≈ 22.7 ms per instance. We find the inference time to be sufficiently low to warrant the real-time and near-real-time functionality of our proposed framework in the aviation domain. Because of its lightweight nature, our proposed methodology could be implemented on the edge computing setup present on the aircraft or ground stations to monitor time-critical anomalies without adding latency to the normal operating procedures.

6 Conclusion and future works

This study presents an artificial intelligence-powered framework for the real-time detection of GNSS interference in ADS-B systems using Convolutional Neural Networks (CNNs) to strengthen protocols in aviation security. The proposed model, based on both CNN and Gated Recurrent Unit (GRU) architectures, is shown to successfully distinguish between interference and authentic GNSS signals, allowing for low-latency processing suitable for high-density airspace conditions. The use of AI-powered predictive analytics supports aviation cybersecurity through the ability to detect potential jamming and spoofing threats before they can affect air traffic operations. Experimental evaluations confirm the proposed model's superiority over traditional approaches through increased detection accuracy, reduced false alarms, and improved robustness of the system. The findings are a step towards creating robust and scalable security solutions for modern air traffic surveillance systems.

Despite the positive results, the work identifies several hurdles, including the class distribution skew in anomaly detection, potential scalability, and the need for thorough validation in real-world settings. Solving these hurdles is vital to improve the model's performance and make it suitable for real-world aviation scenarios. Future work will be focused on addressing the class balance problem identified by this work, owing to its substantial impact on the performance of anomaly detection. Towards enhancing the model's ability to effectively recognize

rare events of GNSS interference episodes, diverse data augmentation techniques will be examined, such as the generation of synthesized anomalies with the use of SMOTE (Synthetic Minority Over-sampling Technique). Additionally, the use of weighted loss functions will enable fairer learning, thus preventing bias towards the major class and making the model more responsive to anomalies in the minority class. All these optimizations are expected to lead to a stronger and more reliable interference detection framework, thus making it more suitable for use in real-world applications for aviation security.

Future research activities will seek to improve the model's robustness through the use of advanced resampling techniques, anomaly-specific loss functions, and reinforcement learning approaches to anomaly detection. Additionally, the use of external flight metadata, including meteorology and pilot performance metrics, holds the promise to further enhance predictive performance and situational awareness. Practical implementation and testing of the model in dynamic air traffic scenarios will be necessary for its performance evaluation under varying conditions. Also, reprocessing pipeline will be extended further to handle class imbalance by synthesizing SMOTE (Synthetic Minority Over-sampling Technique) for creating synthetic anomalies and investigating anomaly-specialized loss functions for optimum detection of rare events. Finally, extending the edge computing capabilities of the framework will enable real-time inference with increased performance, thus enabling widespread adoption in different aviation networks.

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Data availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest All authors declare that they have no conflict of interest. I declare that I have no significant competing financial, professional, or personal interests that might have influenced the performance or presentation of the work described in this manuscript.

Ethical approval This article does not contain any studies with animals performed by any of the authors.

Consent for publication Authors have accepted this article for publication.

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